Supplementary material

Certified threshold decisions from error-bounded river water-quality records over one-way satellite IoT links

Supplementary Table S1: sites of the sealed confirmation cohort

The 35 sites of the second confirmation were selected by metadata only (seed 20261113, at most five per region; the full selection log is results/confirm4/selection.csv). The days are four-parameter days with a record plan in the confirmation run (10,417 in total).

Supplementary Table S1: Sites of the sealed 2022 cohort by region.

Site	Name	State	Days
Gulf and lower Mississippi			
07178200	Bird Ck At State Highway 266 Near Catoosa, Ok	OK	100
07374000	Mississippi River At Baton Rouge, La	LA	324
Mid-Atlantic and Appalachia			
01480870	East Branch Brandywine Creek Below Downingtown, Pa	PA	265
01575900	Chiques Creek Near Marietta, Pa	PA	354
01645762	S F Little Difficult Run Above Mouth Nr Vienna, Va	VA	302
01646305	Dead Run At Whann Avenue Near Mclean, Va	VA	310
01649500	Northeast Branch Anacostia River At Riverdale, Md	MD	340
Mountain West			
09149500	Uncompahgre River At Delta, Co	CO	356
09180500	Colorado River Near Cisco, Ut	UT	307
12301933	Kootenai River Bl Libby Dam Nr Libby Mt	MT	357
13093383	Snake River At Pigeon Cove Nr Twin Falls, Id	ID	361
393259107194801	Colorado R Abv Roaring Fork R At Glenwood Spgs, Co	CO	299
Northeast and New England			
04249000	Oswego River At Lock 7, Oswego Ny	NY	343
Ohio valley and Great Lakes			
03292500	South Fork Beargrass Creek At Louisville, Ky	KY	340
03297900	Floyds Fork Near Pewee Valley, Ky	KY	241
03301900	Fern Creek At Old Bardstown Rd At Louisville, Ky	KY	312
03302060	Salt River At Mouth At West Point, Ky	KY	299
05576100	Lick Creek Near Woodside, Il	IL	303
Pacific coast			
11273400	San Joaquin R Ab Merced R Nr Newman Ca	CA	251
12100490	White River At R Street Near Auburn, Wa	WA	251
14181500	North Santiam River At Niagara, Or	OR	239
14202980	Scoggins Ck Blw Henry Hagg Lake, Nr Gaston, Or	OR	359
14316460	North Umpqua R At Soda Spgs, Nr Toketee Falls, Or	OR	358
Southeast			
0208062765	Roanoke River At Halifax, Nc	NC	349
02203655	South River At Forrest Park Road, At Atlanta, Ga	GA	344
02336300	Peachtree Creek At Atlanta, Ga	GA	330
02336340	Nancy Creek At Johnson Ferry Rd, At Chamblee, Ga	GA	328

Site	Name	State	Days
02336410	Nancy Creek At West Wesley Road, At Atlanta, Ga	GA	333
Texas			
08065350	Trinity Rv Nr Crockett, Tx	TX	264
08188060	San Antonio Rv At Sh 72 Nr Runge, Tx	TX	299
Upper Midwest			
04067500	Menominee River Near Mc Allister, Wi	WI	274
040851385	Fox River At Oil Tank Depot At Green Bay, Wi	WI	350
04087170	Milwaukee River At Mouth At Milwaukee, Wi	WI	148
05474500	Mississippi River At Keokuk, Ia	IA	116
07020850	Mississippi River At Cape Girardeau, Mo	MO	311

Supplementary Table S2: how far each related-work source was read

Access levels used in the manuscript: read in part means the abstract, introduction, method, and text searches were opened; first page means that only the first page was read. No source was read in full. The search log records the 15 searches (all on 2026-11-03, a general web search limited to United States results) and what did not load.

Supplementary Table S2: Reading level of the sources of Table 1.

Source	Level
Albanese et al. (1996)	Introduction, Definitions 3.1 and 3.2, Theorems 3.3 and 3.5, start of Section IV; proofs not read
Boucheron and Salamatian (2000)	First page (abstract and definition)
Brito et al. (2017), Lin et al. (2019)	Introduction, query model, text searched for minimum, maximum, and threshold
Taurone et al. (2024)	Abstract, method, predicate-filtering section
Cheng et al. (2003)	Abstract and introduction; text searched
Diaz-Cacho et al. (2017)	Abstract and introduction
Muñiz-Cuza et al. (2025), Sun et al. (2025)	Abstract and contribution list
Imai and Iri (1988), Deligiannakis et al. (2007)	Read in earlier rounds of the study (level not recorded in the search log)
Lin et al. (2025, Sensors)	Abstract and introduction (the 2019 Energies paper was not opened)

Supplementary Table S3: the first confirmation (version 2 of the method)

The first confirmation tested version 2 of the method (CRC-16 frame, slot-model comparisons, flags that do not use the record) on 31 sealed sites of 2024 (plan docs/CONFIRM3_PLAN.md, results docs/CONFIRM3_RESULTS.md). It is reported here in full because it shaped the revision.

Pairs of values are independent and Gilbert loss at six messages per day. Caveats found after that run: H7 and the configured and record parts of H9 hold by construction; the flags comparator of H5 discarded the record; flags were modelled at 4.75 values per message; and the forgery study of H8 changed body bytes only. After that run its cohort became development data.

Supplementary Table S3: Hypotheses of the first confirmation and their outcome on its 31 sealed sites.

ID	Statement	Limit	Result (95% interval)	Outcome
H1	No wrong decision in any certificate design; no frame round-trip mismatch	0 wrong; 0 mismatches	wrong decisions max 0.000; round trip: 0 mismatches in 30285 blocks	Supported
H2	3b rule minus record only, hard cases at configured thresholds	lower bound at least 0.15	+0.318 (+0.234 to +0.417); +0.300 (+0.223 to +0.390)	Supported
H3	3b rule minus random-day layering, hard cases	lower bound at least 0.05	+0.141 (+0.105 to +0.174); +0.130 (+0.096 to +0.165)	Supported
H4	Record recovered, 3b rule minus record only	lower bound at least −0.12	−0.083 (−0.105 to −0.062); −0.071 (−0.091 to −0.053)	Supported
H5	Unanticipated thresholds, hard cases: 3b rule minus flags, and minus record only	both lower bounds at least 0.10	flags: +0.531 (+0.494 to +0.569); +0.441 (+0.411 to +0.472); record only: +0.260 (+0.191 to +0.344); +0.245 (+0.184 to +0.320)	Supported
H6	Neither send-on-delta nor tolerance x 0.5 dominates the 3b rule	no dominance	send-on-delta/independent: no; tolerance x 0.5/independent: no; send-on-delta/Gilbert: no; tolerance x 0.5/Gilbert: no	Supported
H7	Known limit: flags minus 3b rule, configured thresholds, independent loss	lower bound above 0	+0.043 (+0.025 to +0.068)	Supported
H8	False-certificate day share given one CRC-valid forgery	upper bounds at most 0.06, 0.01, 0.15, 0.02	record, Q = 6: 0.0463 (upper 0.0584, limit 0.06); record, Q = 12: 0.0061 (upper 0.0134, limit 0.01); layered, Q = 6: 0.1050 (upper 0.1197, limit 0.15); layered, Q = 12: 0.0126 (upper 0.0215, limit 0.02)	Not supported
H9	Hybrid 1 minus flags: unanticipated thresholds; configured thresholds; record recovered	alt lower bound at least 0.20; others at least −0.01	alt: +0.355 (+0.297 to +0.418); +0.294 (+0.248 to +0.345); configured: +0.000 (+0.000 to +0.000); +0.000 (+0.000 to +0.000); record: +0.000 (+0.000 to +0.000); +0.000 (+0.000 to +0.000)	Supported

Supplementary note S4: files and reproduction

Everything below is in the project repository; commands run from the Paper3 folder.

- Second confirmation: frozen plan and freeze record docs/CONFIRM4_PLAN.md and docs/FREEZE4.md

(tag confirm4-freeze); data preparation python -I src/p3v3/prepare_cohort4.py; run python -I src/p3v3/confirm4.py data/clean/usgs_new4_2022 confirm4 data/clean/usgs_new4_2022.md5; outputs in results/confirm4 (verdicts4.md, verdicts4.csv, RUN4.log); write-up docs/CONFIRM4_RESULTS.md.
- Development rehearsals with the same code: results/rehearsal4_2023 and results/rehearsal4v2; development record docs/V3_DEV_RESULTS.md and docs/METHOD_V3_PLAN.md; internal reviews docs/REVIEW_FINDINGS_2026-11-09.md and docs/REVIEW_FINDINGS_V3.md.
- First confirmation: docs/CONFIRM3_PLAN.md, docs/FREEZE3.md (tag confirm3-freeze), results/confirm3, docs/CONFIRM3_RESULTS.md.
- Tables and figures of the manuscript: python -I -W ignore paper_tools/make_v1.py.

Supplementary note S5: proofs of Propositions 1 and 2

Setting. A channel-day has readings $x(0),\dots,x(n-1)$. A reading is censored when it lies outside the quantiser range; V is the set of uncensored readings, and every query is defined on V. The server holds for every reading an interval $[\mathrm{lo}(t),\mathrm{hi}(t)]$, a set E of readings known to be censored, and optional global bounds g_{lo}, g_{hi} (from witnesses); it treats S, the complement of E, as uncensored. The knowledge is valid on V when $x(t) \in [\mathrm{lo}(t),\mathrm{hi}(t)]$ for all $t \in V$, $g_{lo} \le x(t) \le g_{hi}$ for all $t \in V$, and $E \cap V = \emptyset$; then $V \subseteq S$. All rules use the effective intervals $\mathrm{lo}' = \max(\mathrm{lo}, g_{lo})$ and $\mathrm{hi}' = \min(\mathrm{hi}, g_{hi})$, which are valid on V when the knowledge is. Under assumption R both ends are moved inward to the reporting grid; because every reading in V lies on the grid, the moved interval still contains it, and the arguments below hold for the moved intervals unchanged.

Lemma 1 (tolerance interval). For $r < 1$ and $x \ge 0$, $\{x : |x-\hat{x}| \le \max(a, rx)\} = [\min(\hat{x} - a, \hat{x}/(1+r)), \max(\hat{x}+a, \hat{x}/(1-r))]$. Proof: the set is the union of $\{|x-\hat{x}| \le a\}$ and $\{|x-\hat{x}| \le rx\}$; both contain $\hat{x}$, the first is $[\hat{x}-a, \hat{x}+a]$, and the second, for $x \ge 0$, is $[\hat{x}/(1+r), \hat{x}/(1-r)]$. Two intervals that share a point have an interval as union, with the extreme endpoints.

Lemma C (cell). Nearest rounding maps x to code c only if $|x - \tilde{x}(c)| \le s/2$, where $\tilde{x}(c)$ is the dequantised value and s the step (on the logarithmic grid of conductance, a factor of $10^{s/2}$).

Proposition 1, part 1 (answers that hold over any superset). Let the knowledge be valid on $V \subseteq S$. (i) Threshold rule "some reading below θ", answer 0: every $t \in S$ has $\mathrm{lo}'(t) \ge \theta$; since $V \subseteq S$ and each $x(t) \ge \mathrm{lo}'(t)$ on V, no reading of V is below θ. The rule "some reading above θ" is symmetric. (ii) Conductance rule, answer 0: every pair of S at the lag is certainly within the fraction; the pairs of V are a subset, and each pair's ratio lies in the image of its intervals (exact for a box), so every pair of V is within the fraction. (iii) Day minimum, lower end: $\min_S \mathrm{lo}' \le \min_V \mathrm{lo}' \le \min_V x$. Day maximum, upper end: $\max_S \mathrm{hi}' \ge \max_V \mathrm{hi}' \ge \max_V x$. None of these uses $S \setminus V$.

Proposition 1, part 2 (answers that need condition B). Condition B: for every $t \in S \setminus V$ there are $u, v \in V$ with $\mathrm{lo}'(t) \le x(u)$ and $\mathrm{hi}'(t) \ge x(v)$. (i) "Some reading below θ", answer 1: some $t \in S$ has $\mathrm{hi}'(t) < \theta$. If $t \in V$, then $x(t) \le \mathrm{hi}'(t) < \theta$. If $t \in S \setminus V$, then $x(v) \le \mathrm{hi}'(t) < \theta$ for some $v \in V$. In both cases a reading of V is below θ. "Above" is symmetric. (ii) Day minimum, upper end: $\min_S \mathrm{hi}'$ is attained at some t. If $t \in V$, then $\min_V x \le x(t) \le \mathrm{hi}'(t)$; if $t \in S \setminus V$, then $\min_V x \le x(v) \le \mathrm{hi}'(t)$. Hence $\min_V x \le \min_S \mathrm{hi}'$, and with part 1 the minimum lies in its interval. The maximum is symmetric. (iii) 7DADM: each daily-maximum interval contains the true daily maximum by (ii) and part 1 (a day with no information contributes the base range, which contains it as well); the mean of seven values lies between the means of the lower and of the upper ends, so

answer 1 (lower mean above the threshold) and answer 0 (upper mean at most the threshold) are correct.

Proposition 1, part 3 (conductance, answer 1). If no conductance reading is in $S \setminus V$, then $S = V$ on that channel, the knowledge is valid on every reading the rule uses, and the rule is the exact image of the box under "all pairs within the fraction" along each chain (a reachable interval propagated by $[0.9a, 1.1b]$ and intersected with the next interval); an empty reachable set means no assignment of the true readings avoids a change, so the answer 1 is correct. With an unrecognised censored conductance reading the chain would pass through a reading outside the domain, and part 3 makes no claim.

Lemma 2 (when condition B holds). Let $t \in S \setminus V$ lie in a certified span whose bounding kept values have uncensored codes c_p, c_q at readings $p, q \in V$. The reconstruction $\hat{x}(t)$ is the linear interpolation of $\tilde{x}_p = \tilde{x}(c_p)$ and $\tilde{x}_q = \tilde{x}(c_q)$ (or equals one of them at the ends of the day), so $\min(\tilde{x}_p, \tilde{x}_q) \le \hat{x}(t) \le \max(\tilde{x}_p, \tilde{x}_q)$. By Lemma 1, $\mathrm{hi}(t) \ge \hat{x}(t) + a$ (linear parameters) or $\hat{x}(t)/(1-r)$ (conductance), and by Lemma C, $x(p) \le \tilde{x}_p + s/2$. The tolerance constants exceed half a step for every parameter (at least 0.2 against at most 0.0025 units; 3% against 0.07% for conductance); therefore, $\mathrm{hi}(t) \ge \min(x(p), x(q))$ and, symmetrically, $\mathrm{lo}(t) \le \max(x(p), x(q))$. Both bounds survive the effective intervals: a witness bound holds on V, so $g_{lo} \le \min(x(p), x(q))$ and $g_{hi} \ge \max(x(p), x(q))$, and the grid move keeps the grid values $x(p), x(q)$ inside. Hence condition B holds with $u, v \in \{p, q\}$. If a bounding kept value has a censored code, its dequantised value is the edge of the range and the argument fails on that side only.

Proposition 2 (exactness without witnesses). Without witnesses, the set of records consistent with the knowledge is the box $\prod_t [\mathrm{lo}(t), \mathrm{hi}(t)]$ over S (taking $S = V$, as when no censored reading is unrecognised). For "some reading below θ": over the box, $\min_t x(t)$ ranges exactly over $[\min \mathrm{lo}, \min \mathrm{hi}]$ (each end attained by setting every reading to its lower or upper end). The query is true on every consistent record iff $\min \mathrm{hi} < \theta$, false on every one iff $\min \mathrm{lo} \ge \theta$, and takes both values otherwise; these are exactly the cases in which the rule answers 1, 0, and -1. "Above" and the extrema are symmetric, and the extremum ends are attained. With a witness, the consistent set is no longer a box: the witness also asserts that some reading lies in its cell. Counterexample: reading 1 in $[3, 10]$, reading 2 in $[4.5, 4.6]$, minimum witness in $[4.0, 4.01]$. Reading 2 cannot be the minimum; therefore, reading 1 lies in $[4.0, 4.01]$ and "some reading above 5" is certainly false, but the box rule (which uses only $x \ge g_{lo}$) answers -1. The rules are therefore sound (Proposition 1) but not exact with witnesses.